

7	(a) (i) (rate of decay) not affected by any external factors or changes in temperature and pressure etc.	B1	[1]
	(ii) two protons and two neutrons	B1	[1]
	(b) (i) (total) mass before decay/on left-hand side is greater than (total) mass on right-hand side/after the decay	M1	
	the difference in mass is released as kinetic energy of the products	A1	[2]
	(may also be some γ radiation) (to conserve mass-energy)		
	(ii) $(6.2 \times 10^6 \times 1.6 \times 10^{-19} =) 9.9(2) \times 10^{-13} \text{ J}$	A1	[1]